06/03

Initiazation of structure variable

struct ABC { int data1; char s1[20]; };

method1 : along with variable declaration.

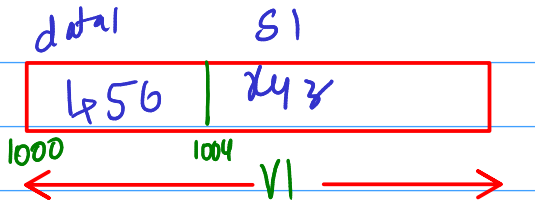
ex :

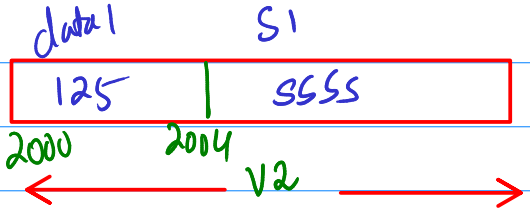
```
struct ABC v={123,"bacd"};
```

method2 : after declaration

```
struct ABC v1,v2;
member by member
        v1.data1=456;
        strcpy(v1.s1,"xyz");

runtime input:
        scanf("%d",&v2.data1);
        scanf("%s",v2.s1);
```





How to copy from one structure variable to another structure variable

method1 : using assignment operator

ex :

```
struct ABC v1={123,"abcd"};
struct ABC v2;
v2=v1;  // data1 value of v1 is copied to data1 of v2
        // s1 of v1 is copied to s1 of v2
```

method2 : using memmove or memcopy

```
memmove(void *dest,void *src,size_t size);
memmove(&v2,&v1,sizeof(v1));
```

method 3: member by member copy

```
v2.data1=v1.data1;
strcpy(v2.s1,v1.s1);
```

Comparing of structure variables
We need to compare member by member

Note : v1==v2 is invalid compile time error

Array of Structures

datatype arraname[size];

struct tagname arrname[size];

ex :

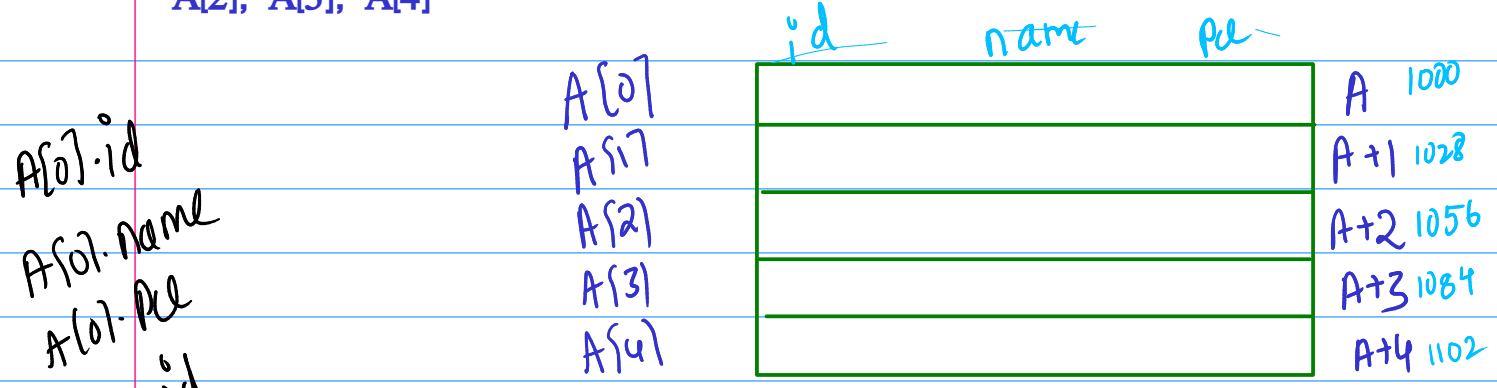
struct Student A[5];

// A is an array 5 struct Student variables

// A[0] is a structure variable

A[1] is a structure variable

A[2], A[3], A[4]



A[0].id
A[0].name
A[0].pcr
A[1].id
A[1].name
A[1].pcr
A[2].id
A[2].name
A[2].pcr

```
for(i=0; i<SIZE; i++)
{ A[i].id;
  A[i].name;
  A[i].pcr;
}
```

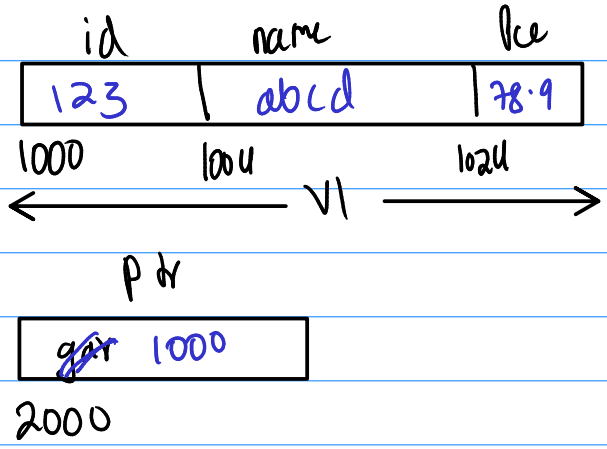
syntax of pointer
datatype *ptrname;

struct tagname *ptrname;
ex : struct Student *ptr;

here size of structure pointer is 8bytes.

```
struct Student v1={123,"abcd",78.9};
struct Student *ptr;

ptr=&v1;
```



*ptr : → fetch 28 byte complete whole record by using record name

we use -> indirection operator to access the members individually

ptr->id;
ptr->name;
ptr->per;